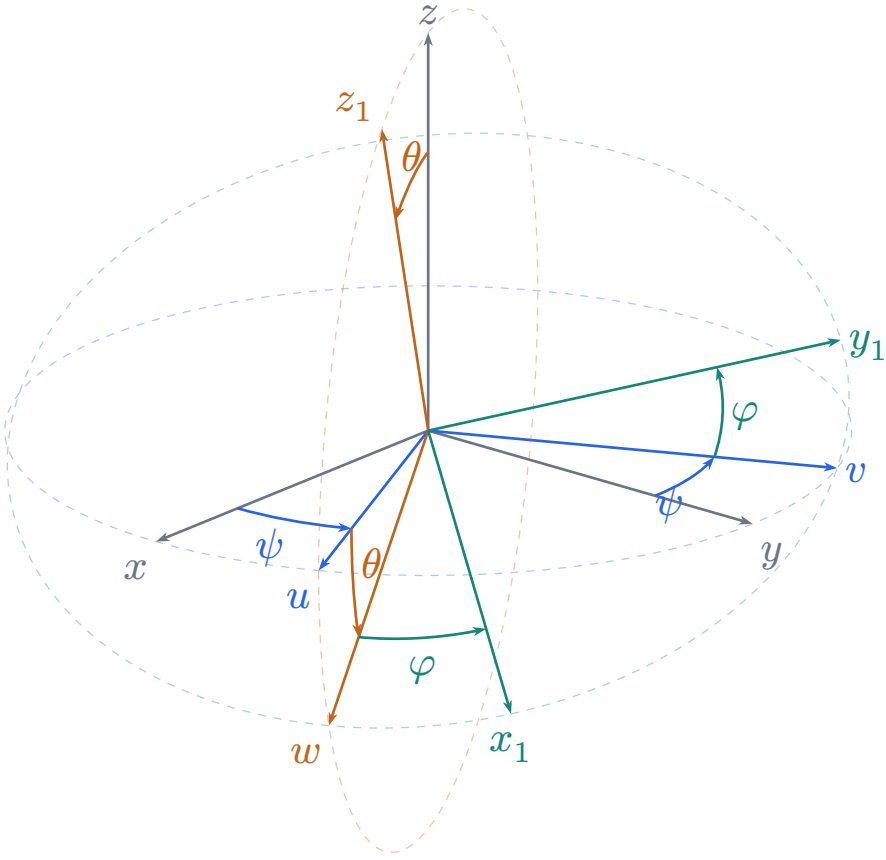
Euler angles: three ordered rotations

Intrinsic Z–Y–Z convention · axes move with the frame



1. Precession ψ Rotate about z

2. Nutation θ Tilt about v

3. Spin φ Rotate about z_1

$$R = R_z(\psi) R_y(\theta) R_z(\varphi)$$

The final basis is (x_1, y_1, z_1).

Changing the rotation order changes the orientation.